



Predicting Severe Asthma Exacerbations

A Genome-Wide Association Study using **Random Forests** Classifiers

Mousheng Xu, PhD

Dec 2025 | GTIIT's 3rd International Symposium Presentation

Self Introduction

Mousheng Xu, PhD

Head of Biosciences, A Private Company In Singapore

- **20+ Years Experience:** Bridging computational biology, statistical genetics, and machine learning.
- **Academic Foundation:**
 - B.S. in Genetics (Wuhan University)
 - M.S. in Biochemistry & Molecular Biology, M.S. in Computer Science (University of Kansas)
 - PhD in Bioinformatics (Boston University; Harvard Medical School)
 - Postdoc in Cancer Genetic Epidemiology (Harvard T.H. Chan School of Public Health).

Key Achievements:

- - Led large-scale GWAS & NGS pipelines.
 - Founded Hubei Vidagen (Genetic Testing).
 - Scientific Breakthrough Award (ATS 2009).

The Clinical Challenge: Severe Asthma Exacerbations



Definition

Severe Asthma Exacerbations: **emergency room visit or a hospitalization** for asthma symptoms.

Prognostic Significance

The ability to predict severe asthma exacerbations would therefore have **direct prognostic significance** and might form the basis for the development of **earlier therapeutic interventions**.

The "Small N, Large P" Problem



Sample Size (N)

127 cases

290 controls

417 total



Standard Method

**Logistic
Regression**

Struggles when predictors (P)
vastly outnumber samples (N).
Cannot easily model complex
interactions.



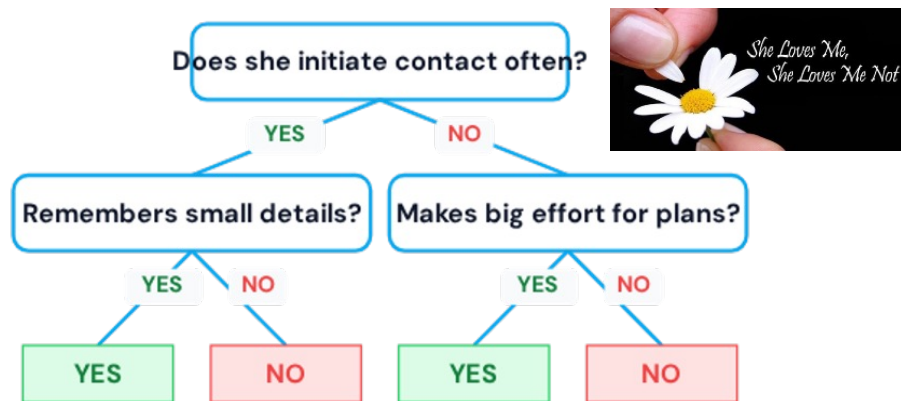
Predictors (P)

> 550,000

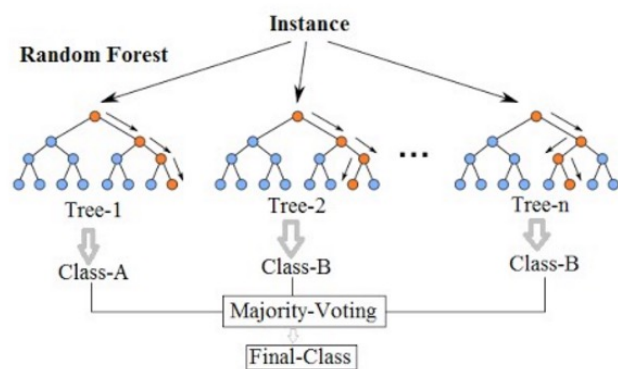
Single Nucleotide
Polymorphisms (SNPs) + Clinical
Traits

Machine Learning With Random Forests

Leveraging ensemble power for robust predictive modeling.



Random Forest Simplified



Key Concepts of Random Forests

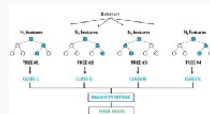
- ✓ A **decision tree** predicts outcomes by splitting data based on feature values.
- ✓ An **ensemble of many decision trees** forms a "forest".
- ✓ Each tree uses **random samples** of the data and considers only **random features** at each split, enhancing diversity.
- ✓ **Voting Mechanism:** The model aggregates predictions by taking the **votes from each decision tree** to determine the final outcome.
- ✓ Handles **nonlinear, high-dimensional data** effectively, crucial for complex datasets.
- ✓ Provides valuable **feature importance** scores, indicating which variables are most predictive.

Key Terminology



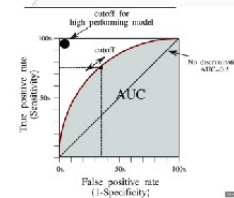
SNP

- **Single Nucleotide Polymorphism**
- A one-letter difference (A/C/G/T) on DNA
- May influence traits or disease risks
- A DNA “mutation” (inappropriate but close)



Importance Score

- Random Forest **predictive power**: contribution to the predictive accuracy
- The higher, the more contribution

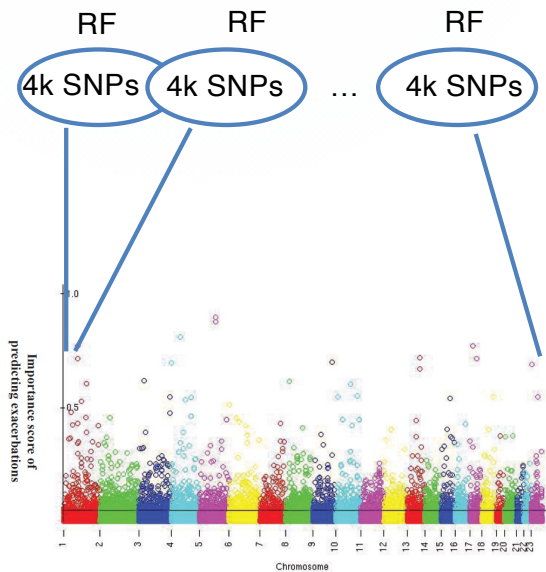


AUC

- **Area Under the Curve**
- A performance metric ranging from 0 to 1.
- Measures the ability to distinguish between classes.
- **0.5 is random guessing**
- **1.0 is perfect prediction.**

Methodology: Feature (SNP) Selection

Step 1: Initial Screen



Step 2: Ranking

Top 4k SNPs
Importance
Score



RF

Reranked 4k SNPs
Importance
Score



Step 3: Prediction

RF

Build final models using
Top N SNPs (10, 40, 160,
320) + 4 Clinical Traits.

Figure 1 The “manhattan plot” of RF importance scores of all the 550k SNPs. X-axis: the SNPs in chromosomal order; Y-axis: the RF importance scores. The black demarcation separates the top 4k SNPs from the rest.

Visualizing the Signal

The RF Importance Landscape

Unlike a traditional GWAS Manhattan plot which shows $-\log(p\text{-values})$, this plot visualizes **Random Forest Importance Scores** across the genome.

The "peaks" represent SNPs that contribute most significantly to classifying a patient as an "Exacerbator" vs. "Non-Exacerbator".

We identified a clear signal distinguishing the top predictive SNPs from the genomic background noise.

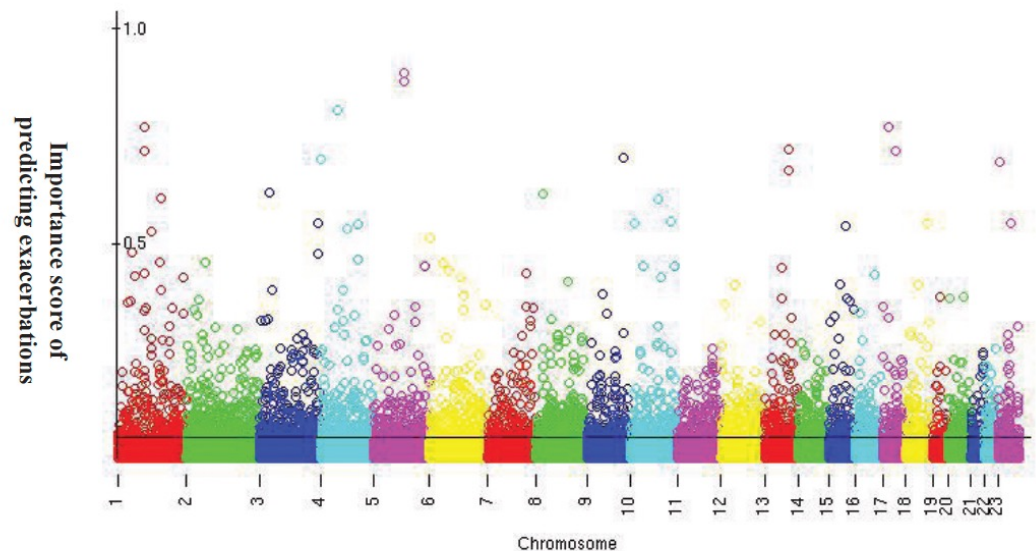
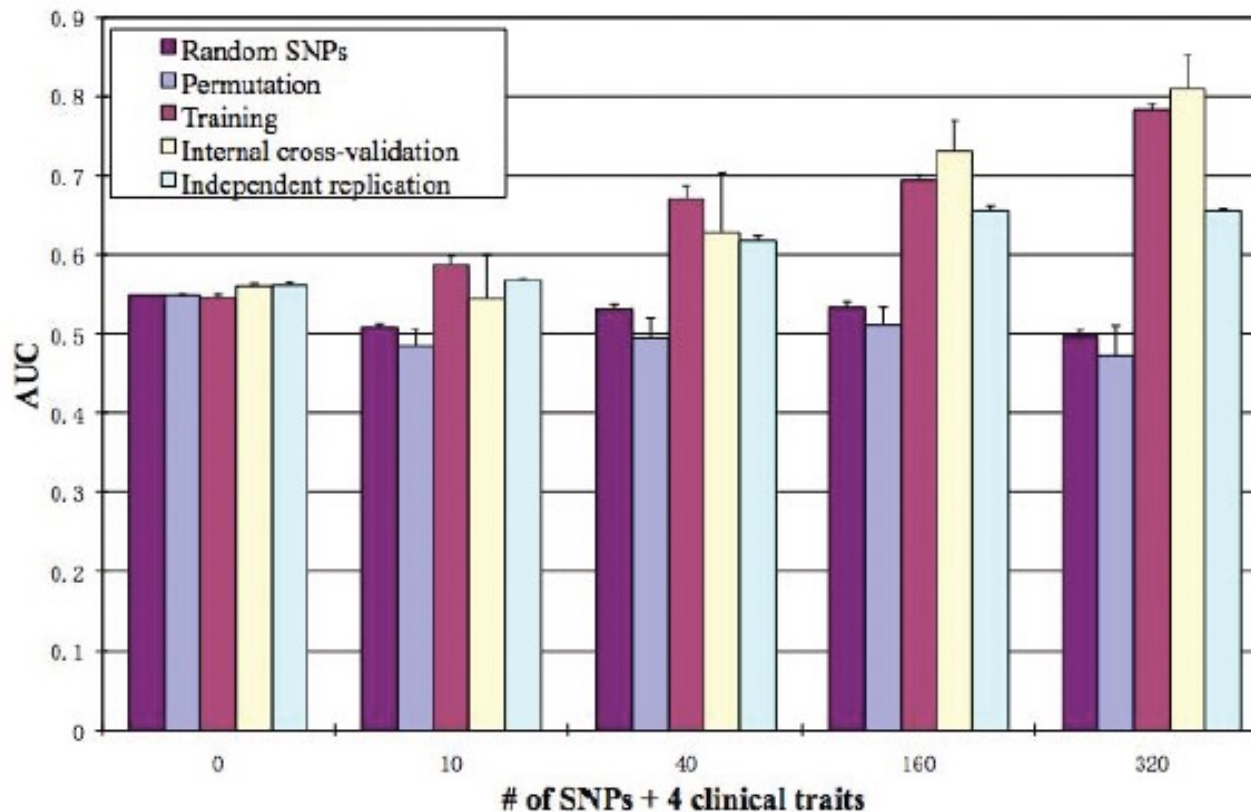


Figure 1 The "manhattan plot" of RF importance scores of all the 550k SNPs. X-axis: the SNPs in chromosomal order; Y-axis: the RF importance scores. The black demarcation separates the top 4k SNPs from the rest.

4 months after publication, Editor:

Genetics Improves Prediction

Area Under the Curve (AUC) in Independent Replication Cohort



Key Finding: Adding genetic markers significantly improved prediction over clinical history alone. The model plateaued at 160 SNPs, suggesting a robust genetic signature for exacerbation risk.

Validation & Impact



Rigorous Validation

We did not just test on the training data.

- **Permutation Testing:** Confirmed results were not due to chance.
- **Independent Replication:** The model was tested on a separate cohort of 164 subjects, maintaining an AUC of 0.66.



Recognition

This work was highlighted for its innovative approach:

- **ATS 2009:** Featured at the "**Scientific Breakthroughs of the Year**" symposium.
- **Impact:** Categorized as "**Highly Accessed**" (1,120 accesses 4 months after publication) by BMC Medical Genetics, influencing future ML-GWAS integration studies.

Selected Publications: GWAS & NGS

Genome-Wide Association Studies (GWAS)

- Handedness GWAS (Meta-analysis) — Nat Hum Behav (2020)
- Basal Cell Carcinoma — PLoS One (2011)
- Nevus Count / Melanoma Risk (NID1) — Hum Mol Genet (2011, Co-first)
- Cutaneous BCC & SCC — Hum Mol Genet (2011)
- Pancreatic Adenocarcinoma Survival — Gut (2012)
- Severe Asthma Exacerbation (RF GWAS) — BMC Med Genet (2011)
- eQTL Mapping — Hum Mol Genet (2010)
- ABO & Pancreatic Cancer — Cancer Epidemiol Biomarkers Prev (2010)
- Blood Type & Breast Cancer — Int J Cancer (2011)

NGS & Functional Genomics

- Synthetic transcription elongation factors — Science (2017)
- YAP1 → USP31 → NFκB (Sarcoma) — Cancer Research (2018)
- TRIM24 degrader — Nat Chem Biol (2018)
- MYC dependency in ovarian cancer — eLife (2018)
- Super-enhancer BCL2 mechanism — Cancer Cell (2018)
- Warfarin pharmacogenetics — Pharmacogenomics (2015)

ML Expertise & Industry Experience

Functional Genomics & Machine Learning

- ENCODE Pilot Project — Nature (2007)
- Regulatory polymorphisms / eQTL — Genomics (2008)
- Lgl1 regulation during alveolarization — Pediatric Research (2010)
- RF model for asthma exacerbation — BMC Med Genet (2011)
- LSTM Stock Prediction — Kaggle Tutorial (2018)
- Biological information retrieval — ACM CIKM (1998)

Industry Highlights

- Vidagen (Founder): Genetic testing; developed dozens of products; ~ a dozen patents. (Awards: Hundred, Yellow Crane, 3551 Talent Programs).
- Booz Allen Hamilton (Sr. Data Scientist):
 - Genomic data preparation for Million Veteran Program (MVP).
 - Natural Language Processing (NLP) for NIDDK.
- D2M Biotherapeutics: Causal gene identification for cancers and immune diseases (11M+ MR results).
- Head of Biosciences (Singapore): Quantum-physics for biomedical applications.

Vision: Integrating Computational Analysis into BFE

My goal is to establish a functional lab at GTIIT that bridges the gap between biological engineering and data science.



Curriculum

Introducing practical Machine Learning and Bioinformatics training specifically designed for BFE students to modernize their skillset.



Research

Developing explainable AI pipelines for complex food and biological data, enabling data-driven discoveries in biotech.



Innovation

Translating large-scale "Big Data" into actionable engineering insights for industrial applications.

Thank You

Questions & Discussion



Mousheng Xu, PhD



moushengxu@gmail.com